Akoizo Original Practice - Form 2 - Answer key

Division B | Elastic Launched Gliders | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) A glider has glide ratio 6 and airspeed 3.0 m/s. What is its sink rate?

A. 2.0 m/s

B. 0.20 m/s

C. 0.50 m/s

D. 18 m/s

Correct: C

2. (2 points) Which is a required flight log parameter?

Unit: The flight log and competition day.
Worked-case data: A team has five official flights in 6 minutes. After a trim flight, it plans official flights. Its first three are 21, 23, and 8 s (a beam hit). How should it use the remaining two?
Separate investigation observations:
Flight 4: Peak 9 m, time 12 s.
Flight 5: Peak 4 m, time 13 s.
Flight 6: Peak 9 m, time 24 s after reducing rudder offset.

A. Approximate relaxed elastic length

B. Glider color

C. Room temperature

D. Wing camber

Correct: A

3. (2 points) Why reinforce the elastic hook area?

Unit: Construction: light, stiff, and legal.
Worked-case data: A glider weighs 3.4 g and its CG is behind the target. What should the team do?
Separate investigation observations:
Flights 1-4: 22-24 s each.
Flight 5: Hard landing on nose.
Flights 6-8: 9-12 s; glider veers right at launch.

A. It carries the large launch force

B. It adds lift

C. It reduces drag

D. It is required by rule

Correct: A

4. (2 points) Which change best lowers peak height for a low ceiling?

Unit: Launch energy: from stretched elastic to height.
Worked-case data: An elastic with k = 20 N/m is stretched 0.30 m. Half the energy reaches a 6.0 g glider. Find launch speed and ideal height.
Separate investigation observations:
Pull 25 cm: Peak 6 m; flight 19 s.
Pull 35 cm: Peak 9 m; flight 26 s.
Pull 45 cm: Hit 10 m ceiling; flight 15 s.

A. Reduce the pull distance

B. Use a thicker elastic

C. Remove ballast

D. Launch from a chair

Correct: A

5. (4 points) Why does consistency matter more than one long flight?

Unit: Rules, scoring, and planning five flights.
Worked-case data: A team with a complete log flies official flights of 22.4, 18.0, 25.1, 9.6, and 24.0 s. What is the final score and tiebreaker?
Separate investigation observations:
Glider A: Flights: 35, 12, 11, 10, 9 s.
Glider B: Flights: 24, 23, 22, 21, 20 s.
Both: Complete flight log.

answer: The final score adds three flights, so three good flights beat one excellent flight and several poor ones. (4 points).

6. (4 points) Why watch other flights before launching?

Unit: Launch technique and the ceiling.
Worked-case data: In practice, a 35 cm pull reaches 9 m. Competition ceiling is 7.5 m. Estimate a starting pull, assuming height scales with energy.
Separate investigation observations:
Practice room: 12 m ceiling; best pull 40 cm.
Competition room: 6 m ceiling with beams at 5.5 m.
First trim flight: 40 cm pull; glider hit a beam.

answer: To find drafts and obstacles like beams and lights, and choose a launch spot that avoids them. (4 points).

7. (4 points) Why test hand glides before elastic launches?

Unit: Trim: the transition from climb to glide.
Worked-case data: A glider climbs well but loops over at the top and then dives. Suggest a fix.
Separate investigation observations:
Launch at 0° bank: Glider loops over and dives; 8 s.
Launch at 30° bank: Climbs in a curve, rolls level; 22 s.
Launch at 70° bank: Climbs low and spirals; 12 s.

answer: Hand glides show pitch trim at glide speed safely, without the stresses and confusion of a full launch. (4 points).

8. (4 points) What happens if a glider is trimmed too slow?

Unit: Glide physics: sink rate and glide ratio.
Worked-case data: A glider reaches 9.0 m and glides at 3.2 m/s with glide ratio 8. Estimate glide time.
Separate investigation observations:
Glider A: Peak 10 m; sink 0.55 m/s.
Glider B: Peak 8 m; sink 0.35 m/s.
Question: Which glides longer, ignoring climb time?

answer: It approaches stall, losing lift suddenly and dropping, often in a series of stalls. (4 points).

9. (4 points) Why is a second trimmed glider valuable?

Unit: The flight log and competition day.
Worked-case data: A team has five official flights in 6 minutes. After a trim flight, it plans official flights. Its first three are 21, 23, and 8 s (a beam hit). How should it use the remaining two?
Separate investigation observations:
Flight 4: Peak 9 m, time 12 s.
Flight 5: Peak 4 m, time 13 s.
Flight 6: Peak 9 m, time 24 s after reducing rudder offset.

answer: Teams get no extra time for repairs; a ready backup keeps official flights going if one glider breaks. (4 points).

10. (4 points) Why must long wings stay stiff?

Unit: Construction: light, stiff, and legal.
Worked-case data: A glider weighs 3.4 g and its CG is behind the target. What should the team do?
Separate investigation observations:
Flights 1-4: 22-24 s each.
Flight 5: Hard landing on nose.
Flights 6-8: 9-12 s; glider veers right at launch.

answer: Flexing at launch speed causes flutter and erratic climbs, and bending changes the glide trim. (4 points).

11. (6 points) Solve this case and explain your reasoning: An elastic with k = 20 N/m is stretched 0.30 m. Half the energy reaches a 6.0 g glider. Find launch speed and ideal height.

reasoning: Energy stored = ½ x 20 x 0.30^2 = 0.90 J; to glider: 0.45 J.
v = sqrt (2E ÷ m) = sqrt (2 x 0.45 ÷ 0.006) about 12 m/s.
Ideal height = E ÷ (mg) = 0.45 ÷ (0.006 x 9.8) about 7.7 m before drag losses. (3 points).

conclusion: About 12 m/s at launch; drag keeps the real peak below 7.7 m. (3 points).

12. (2 points) Which glider gives the higher final score?

Supplied synthetic observations:
Glider A: Flights: 35, 12, 11, 10, 9 s.
Glider B: Flights: 24, 23, 22, 21, 20 s.
Both: Complete flight log.

A. Glider B

B. Glider A

C. They tie

Correct: A

13. (2 points) Define “Launch angle” in the context of Elastic Launched Gliders.

answer: Angle above horizontal at release. (2 points).

14. (2 points) Define “Transition” in the context of Elastic Launched Gliders.

answer: Change from fast climb to slow glide. (2 points).

15. (2 points) Define “Sink rate” in the context of Elastic Launched Gliders.

answer: Vertical descent speed. (2 points).

16. (2 points) Define “Relaxed elastic length” in the context of Elastic Launched Gliders.

answer: Elastic length when not stretched; required parameter. (2 points).

17. (2 points) Define “Blunt nose” in the context of Elastic Launched Gliders.

answer: Rounded nose required for safety. (2 points).

18. (2 points) Define “Gravitational potential energy” in the context of Elastic Launched Gliders.

answer: mgh, energy of height. (2 points).